The empirical column recurrences for the table OEIS A234161

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Abstract

OEIS A234161 is a two-dimensional table $T(n,k)$, and it carries a block of empirical recurrences contributed by R. H. Hardin — one for each of its first few columns. They are true. Column k of the table is an ordinary fixed-width array count, so its rows are the vertices of a finite digraph and $T(n,k)$ is a number of walks in it; being a walk count the column is C-finite, and whether a given recurrence annihilates it is decided by a finite exact computation, with the Cayley–Hamilton theorem bounding the work. This note settles the columns $k = 1, 2$.

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1 The table and the conjectures

OEIS A234161 is "T(n,k)=Number of (n+1)X(k+1) 0..3 arrays with every 2X2 subblock having the absolute values of all six edge and diagonal differences no larger than 1.". It has offset 1, is read by antidiagonals, and begins

$$46, 226, 226, 1108, 2210, 1108, 5446, 21428, \ \ldots$$

The entry carries the following block:

Empirical for column k:
k=1: a(n) = 7*a(n-1) -9*a(n-2) -6*a(n-3)
k=2: a(n) = 16*a(n-1) -59*a(n-2) -31*a(n-3) +118*a(n-4) +35*a(n-5) -10*a(n-6)
k=3: [order 11]
k=4: [order 26]
k=5: [order 53]

As of the "Last modified" line on the live entry (Jun 02 2025 09:14:22, revision 6) these are still recorded as empirical, and nothing on the entry records any of them as settled.

2 A column of the table is a walk count

Fix k. The entry defines $T(n,k)$ as the number of arrays of a shape in which one dimension is n and the other is determined by k, subject to a condition imposed on every 2×2 subblock — a condition local to two consecutive rows and two consecutive columns. Substituting the fixed value of k into the entry's own wording turns the definition into an ordinary one-parameter array count, and for such a count the rows are the vertices of a finite digraph G_k: $r \to s$ is an edge exactly when placing s under r satisfies the condition in every column pair. An admissible array is then a walk, so with M_k the adjacency matrix of G_k,

$$T(n,k) \;=\; c_k\, \mathbf{1}^\top M_k^n \mathbf{1}$$

for the constant c_k that the entry's name supplies (a factor such as $\frac{1}{2}$ or $\frac{1}{4}$ where the name says so, and 1 otherwise).

Lemma 1. *Let $q(t) = t^r - \sum_i c_i t^{r-i}$ be the characteristic polynomial of a conjectured recurrence of order r for column k. Then for every $n \geq r$,*

$$T(n,k) - \sum_i c_i\, T(n-i,k) \;=\; c_k\, \mathbf{1}^\top M_k^{n-r} q(M_k)\mathbf{1},$$

so the recurrence holds from some point on if and only if $\mathbf{1}^\top M_k^j q(M_k)\mathbf{1} = 0$ for every $j \geq 0$; and by the Cayley–Hamilton theorem it is enough to check $j < S_k$, where S_k is the number of vertices of G_k.

Proof. Substitute the walk expression and factor M_k^{n-r} out of $M_k^n - \sum_i c_i M_k^{n-i}$; the scalar c_k is nonzero. For the bound, $M_k^{S_k}$ is an integer combination of $I, M_k, \ldots, M_k^{S_k-1}$, so each $\mathbf{1}^\top M_k^j q(M_k)\mathbf{1}$ with $j \geq S_k$ repeats that combination of earlier values. $\square$

3 The computation, column by column

column k	order	vertices S_k	proved for	terms checked
1	3	16	$n > 3$	8
2	6	64	$n > 6$	9

In each case the vector $w = q(M_k)\mathbf{1}$ was formed by r matrix–vector products in exact integer arithmetic and $\mathbf{1}^\top M_k^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; every one of those integers vanishes from the stated point onwards and the run continued until the working vector was identically zero, which settles all larger j at once.

Column $k = 1$

The sequence is "Number of (n+1)X(2) 0..3 arrays with every 2X2 subblock having the absolute values of all six edge and diagonal differences no larger than 1.". Its rows are the vertices of a digraph on $S = 16$ vertices, edges being the admissible consecutive pairs, and the count is $\mathbf{1}^\top M^n \mathbf{1}$ up to the constant factor in the entry's name. The conjectured recurrence

$$a(n) \;=\; 7\,a(n-1) - 9\,a(n-2) - 6\,a(n-3)$$

holds for every $n > 3$.

Column $k = 2$

The sequence is "Number of (n+1)X(3) 0..3 arrays with every 2X2 subblock having the absolute values of all six edge and diagonal differences no larger than 1.". Its rows are the vertices of a digraph on $S = 64$ vertices, edges being the admissible consecutive pairs, and the count is $\mathbf{1}^\top M^n \mathbf{1}$ up to the constant factor in the entry's name. The conjectured recurrence

$$a(n) \;=\; 16\,a(n-1) - 59\,a(n-2) - 31\,a(n-3) + 118\,a(n-4) + 35\,a(n-5) - 10\,a(n-6)$$

holds for every $n > 6$.

4 Verification

Three checks, each independent of the algebra above.

First, each column model was compared against the entry's own published values for that column, taken both from the antidiagonal DATA and from the "Table starts" block, and agrees at every available term. This is what ties a model to the table: substituting k into the entry's wording is a reading of English, and a wrong reading gives a different digraph and different counts. The values involved are large — column entries here run to sixteen digits and beyond — so agreement across the available terms is not something a wrong model achieves.

Second, each conjectured recurrence was evaluated on those published column values in exact integer arithmetic, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic rather than on a sampled prefix.

Remark 2. A table row is a column of the transposed table, so the same argument settles the entry's row conjectures whenever the transposed shape is itself of the form treated here. Only the columns the entry states explicitly, and for which enough published terms exist to run the first check above, are claimed.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A234161.

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.